Problem 1: Express the following matrix equation as a system of linear equations:

$$\begin{bmatrix} 3 & -1 & 2 \\ 4 & 3 & 7 \\ -2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

Solution

The given matrix equation can be written as a system of linear equations by interpreting each row of the matrix multiplication:

$$\begin{aligned} 3x_1 - x_2 + 2x_3 &= 2, \\ 4x_1 + 3x_2 + 7x_3 &= -1, \\ -2x_1 + x_2 + 5x_3 &= 4. \end{aligned}$$

Thus, the system of linear equations is:

$$\begin{cases} 3x_1 - x_2 + 2x_3 = 2, \\ 4x_1 + 3x_2 + 7x_3 = -1, \\ -2x_1 + x_2 + 5x_3 = 4. \end{cases}$$

Problem 2: Show that if A is a $p \times q$ matrix, and $A(BA)$ is defined, then B is a $q \times p$ matrix.

Solution

Let B be a matrix of order $m \times n$. For BA to be defined, the number of columns of B must equal the number of rows of A. Thus, we have:

$$n = p,$$

which implies that B is a matrix of order $m \times p$.
By the matrix multiplication rule, the order of BA is given by:

$$\text{number of rows of } B \times \text{number of columns of } A.$$

This implies that the matrix BA has order $m \times q$.
Now, for $A(BA)$ to be defined, the number of columns of A must equal the number of rows of BA. This gives:

$$q = m.$$

Hence, substituting $m = q$ into the earlier result for B, we conclude that B is a matrix of order $q \times p$.

Problem 3: Find matrices A, x, and b that express the given system of linear equations as a single matrix equation $Ax = b$.

$$\begin{aligned} 4x_1 - 3x_3 + x_4 &= 1, \\ 5x_1 + x_2 - 8x_4 &= 3, \\ 2x_1 - 5x_2 + 9x_3 - x_4 &= 0, \\ 3x_2 - x_3 + 7x_4 &= 2. \end{aligned}$$

Solution

The given system of linear equations can be written in the form $Ax = b$, where:

$$A = \begin{bmatrix} 4 & 0 & -3 & 1 \\ 5 & 1 & 0 & -8 \\ 2 & -5 & 9 & -1 \\ 0 & 3 & -1 & 7 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}.$$

Thus, the matrix equation representing the system is:

$$\begin{bmatrix} 4 & 0 & -3 & 1 \\ 5 & 1 & 0 & -8 \\ 2 & -5 & 9 & -1 \\ 0 & 3 & -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}.$$

Problem 4: If A and B are two square matrices of the same size, then find the condition such that:

$$(A + B)^2 = A^2 + B^2 + 2AB.$$

Solution

To determine the condition, we expand $(A + B)^2$ using matrix multiplication:

$$(A + B)^2 = (A + B)(A + B).$$

Expanding this:

$$(A + B)^2 = A^2 + AB + BA + B^2.$$

For $(A + B)^2 = A^2 + B^2 + 2AB$ to hold, we must have:

$$AB + BA = 2AB.$$

Simplifying this equation:

$$BA = AB.$$

Thus, the condition is that A and B must commute, i.e., $AB = BA$.

Problem 5: Let $D = \text{diag}(d_1, d_2, \ldots, d_m)$, and A is an $m \times n$ matrix. By showing appropriate derivation/working, find the rule for pre-multiplication of A by matrix D.

Solution

The matrix $D = \text{diag}(d_1, d_2, \ldots, d_m)$ is a diagonal $m \times m$ matrix with $d_1, d_2, \ldots, d_m$ on the diagonal and zero elsewhere:

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_m \end{bmatrix}.$$

Let A be an $m \times n$ matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The pre-multiplication DA is given below:

$$DA = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_m \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The result is:

$$DA = \begin{bmatrix} d_1 a_{11} & d_1 a_{12} & \cdots & d_1 a_{1n} \\ d_2 a_{21} & d_2 a_{22} & \cdots & d_2 a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_m a_{m1} & d_m a_{m2} & \cdots & d_m a_{mn} \end{bmatrix}.$$

Each row i of A is scaled by the corresponding diagonal entry d_i of D.

Problem 6: Let $D = \text{diag}(d_1, d_2, \ldots, d_m)$, and A is an $n \times m$ matrix. By showing appropriate derivation/working, find the rule for post-multiplication of A by matrix D.

Solution

The matrix $D = \text{diag}(d_1, d_2, \ldots, d_m)$ is a diagonal $m \times m$ matrix with $d_1, d_2, \ldots, d_m$ on the diagonal and zero elsewhere:

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_m \end{bmatrix}.$$

Let A be an $n \times m$ matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}.$$

The post-multiplication AD is given below:

$$AD = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_m \end{bmatrix}.$$

The result is:

$$AD = \begin{bmatrix} a_{11}d_1 & a_{12}d_2 & \cdots & a_{1m}d_n \\ a_{21}d_1 & a_{22}d_2 & \cdots & a_{2m}d_n \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}d_1 & a_{n2}d_2 & \cdots & a_{nm}d_n \end{bmatrix}.$$

In general, post-multiplication of A by D scales each column j of A by the corresponding diagonal entry d_j in D.

Problem 7: Suppose that A and B are 6×8 matrices, and C is a 3×6 matrix. Then, what is the size of the matrix $C(A + B)$?

Solution

Since A and B are both 6×8 matrices, their sum $A + B$ is also a 6×8 matrix. Matrix multiplication $C(A + B)$ is defined only if the number of columns in C matches the number of rows in $A + B$, which holds true in the given case as $6 = 6$. The resulting matrix will have size:

$$\text{Number of rows from } C \times \text{Number of columns from } A + B = 3 \times 8.$$

Problem 8: Suppose that A and B are 4×5 matrices, and C is a 5×4 matrix. Then, what is the size of the matrix $C(A + B)$?

Solution

Since A and B are both 4×5 matrices, their sum $A + B$ is also a 4×5 matrix. Matrix multiplication $C(A + B)$ is defined only if the number of columns in C matches the number of rows in $A + B$, which holds true in the given case as $4 = 4$. The resulting matrix will have size:

$$\text{Number of rows from } C \times \text{Number of columns from } A + B = 5 \times 5.$$

Problem 9: Find the augmented matrix for the following system of linear equations:

$$\begin{aligned} x_1 - 2x_2 + x_3 - 4x_4 &= 1, \\ x_1 + 3x_2 + 7x_3 + 2x_4 &= 2, \\ x_1 - 12x_2 - 11x_3 - 16x_4 &= 5. \end{aligned}$$

Solution

The given system of linear equations can be written as an augmented matrix by extracting the coefficients of x_1, x_2, x_3, x_4 and placing the constants on the right as the last column.
The augmented matrix is:

$$\left[\begin{array}{cccc|c} 1 & -2 & 1 & -4 & 1 \\ 1 & 3 & 7 & 2 & 2 \\ 1 & -12 & -11 & -16 & 5 \end{array}\right].$$

Problem 10: Show that if AB and BA are both defined, then AB and BA are square matrices.

Solution

Let A be an $m \times n$ matrix and B be an $p \times q$ matrix.
For AB to be defined,

$$\text{the number of columns of } A = \text{the number of rows of } B$$

implies $p = n$.
Similarly, for BA to be defined,

$$\text{the number of columns of } B = \text{the number of rows of } A$$

implies $q = m$. And hence B must be of size $n \times m$.

The order of product of two matrices is defined as:

$$\text{the number of rows of left matrix} \times \text{the number of columns of the right matrix.}$$

Which gives us that AB is a matrix of size $m \times m$ and BA is a matrix of size $n \times n$.
Thus, both AB and BA are square matrices.